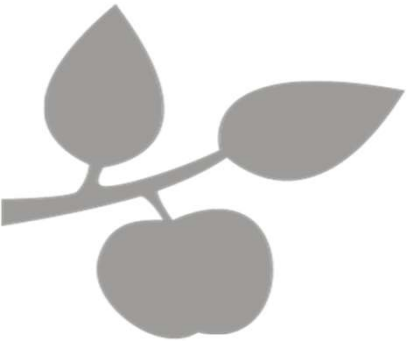




DM583

Data Mining

Probability and Density Estimation

PART 1

Review of Basic Probability and Probability Distributions

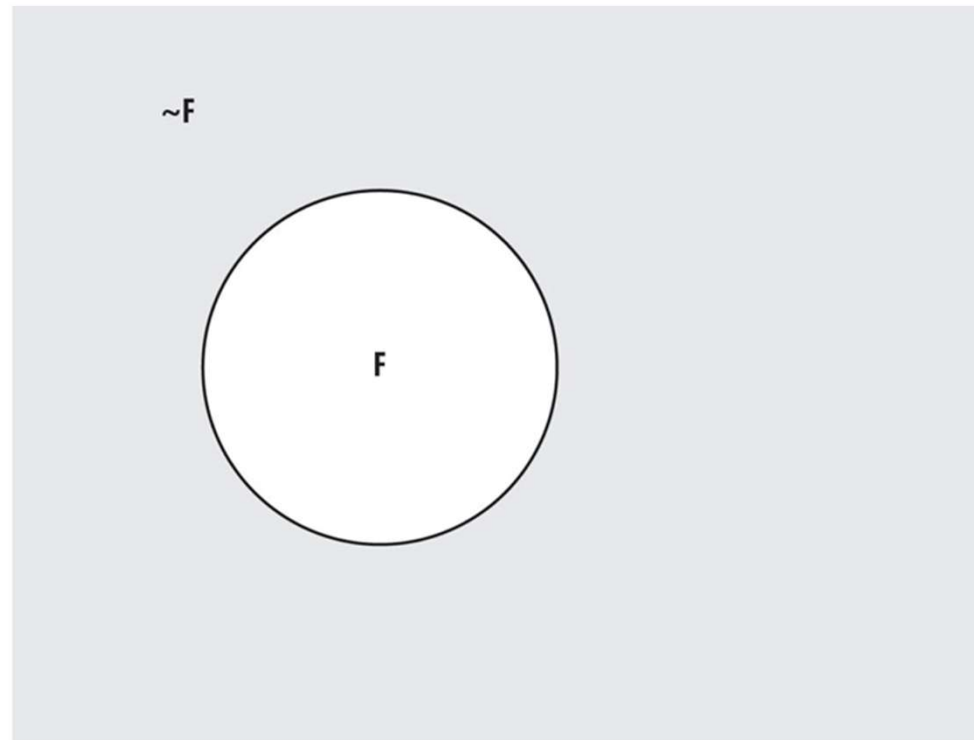
Definition of Probability

- ***Trial*** – a process that produces a result subject to uncertainty
 - It is also called a (statistical) **experiment**
 - For example: 'rolling dice once'
- ***Sample Space*** – a set of all possible outcomes of a trial
 - For example: $S = \{1,2,3,4,5,6\}$
 - Each of these possible outcomes is an **element** (or **member**) of the sample space
- ***Event*** – a subset of the sample space
 - The subset contains the elements for which a given proposition is true
 - For example: 'odd number rolled' $E = \{1,3,5\}$
- ***Probability of Event***: $p(E) = |E|/|S|$

➤ Taken from Christopher M. Bishop, "Pattern Recognition and Machine Learning"

$$p(F) + p(\sim F) = 1$$

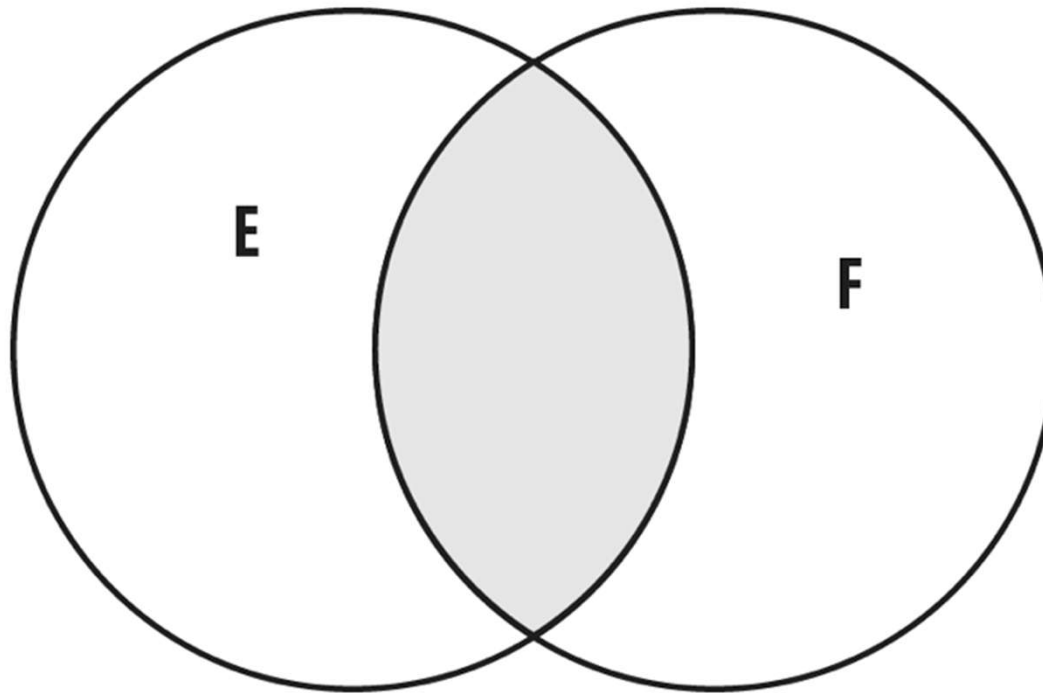
Event complement



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Intersection of Events

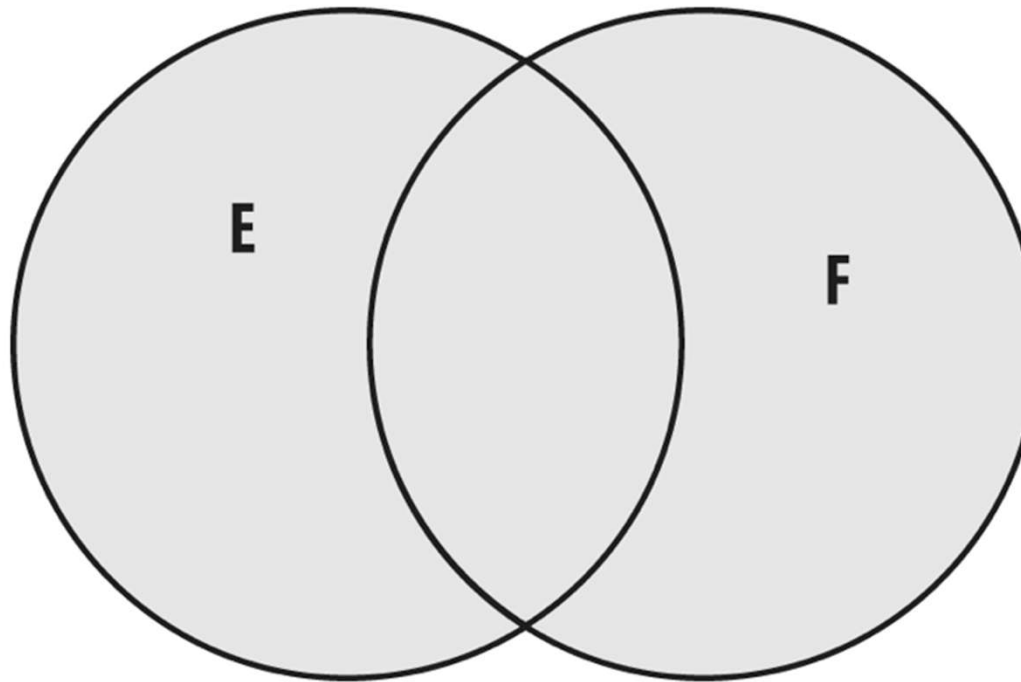
$$p(E \cap F)$$



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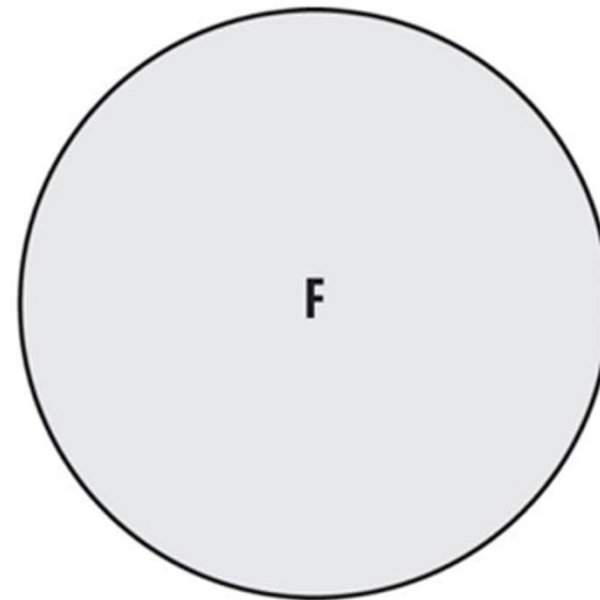
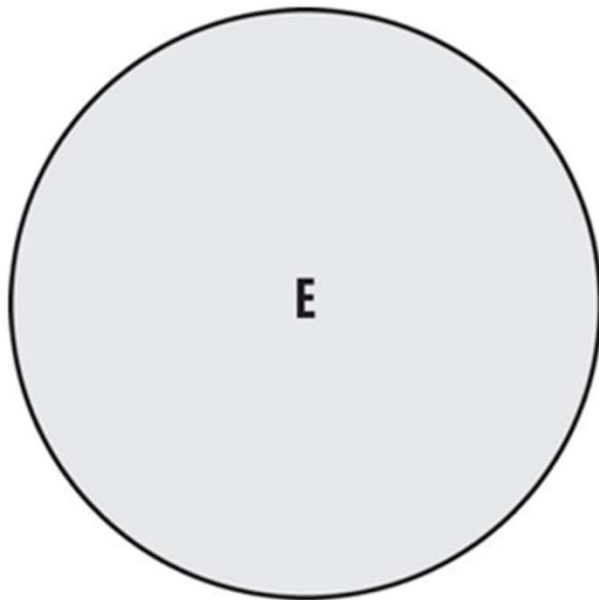
Union of Events

$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$



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Mutually Exclusive Events

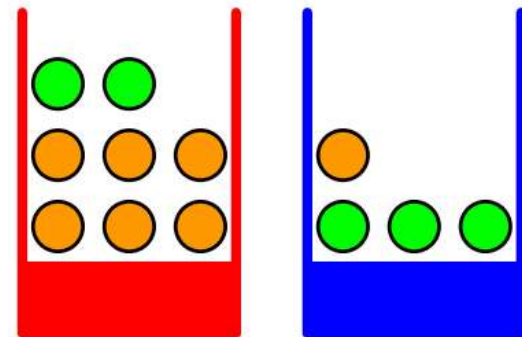


$$p(E \cap F) = 0$$

$$p(E \cup F) = p(E) + p(F)$$

Definition of Probability

- To begin with, we shall define the probability of an event to be the fraction of times that event occurs out of the total number of trials, in the limit that the total number of trials goes to infinity
- Let's look at the following simple experiment:
 - Two boxes, red (r) and blue (b)
 - Each box contains either apples (a) or oranges (o)
 - The box that will be chosen is a **random variable** B . It takes the values r or b
 - The fruit sampled out of the selected box is another **random variable** F and takes the values a or o . These values are *uncertain*



➤ Taken from Christopher M. Bishop, "Pattern Recognition and Machine Learning"

Probability and Random Variables

- Let's assume we select the blue box 60% of the time
- Repeating the box selection infinite number of times, we say the probability of selecting the red box is 4/10 and the blue box 6/10
- We formally write this as follows:

$$p(B = r) = \frac{4}{10} \quad p(B = b) = \frac{6}{10}$$

- Note that, by definition, probabilities must lie in the interval $[0, 1]$
- Also, if events are *mutually exclusive* and *include all possible outcomes*, then we see that the **probabilities for those events must sum to one**

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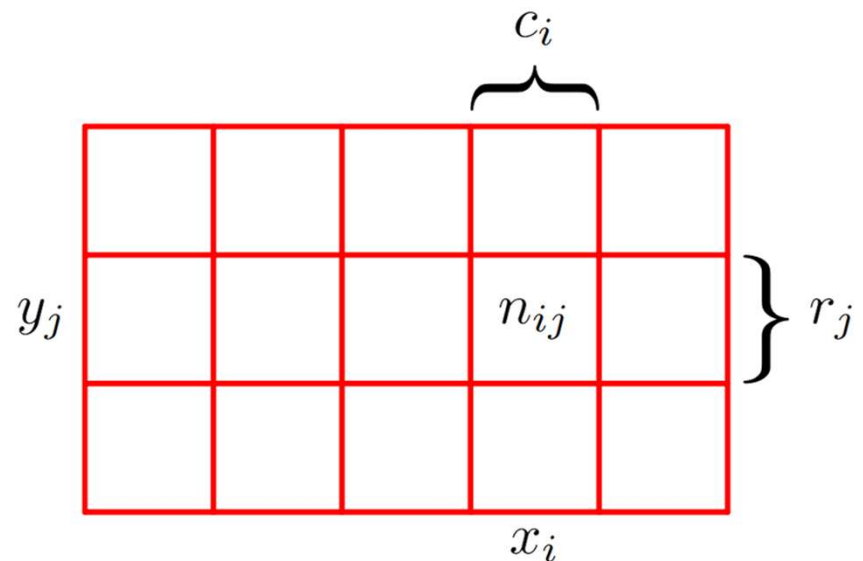
Joint Probability Distributions

- We can now ask questions such as:
 - *“What is the probability that the selection procedure will pick an apple?”*
 - *“Given that we have picked an orange, what is the probability that the box we chose it from was the blue one?”*
- To answer such questions, we need the elementary rules of probability:
 - the **sum rule**
 - the **product rule**

➤ Taken from Christopher M. Bishop, "Pattern Recognition and Machine Learning"

Joint Probability Distributions

- Consider we have two random variables X and Y
 - X can take any of the values x_i where $i = 1, \dots, M$
 - Y can take the values y_j where $j = 1, \dots, L$
- Consider a total of N trials
 - The number of such trials in which $X = x_i$ and $Y = y_j$ is n_{ij}
 - The number of trials in which X takes the value x_i is c_i
 - The number of trials in which Y takes the value y_j is r_j
 - The sum of all n_{ij} values is N



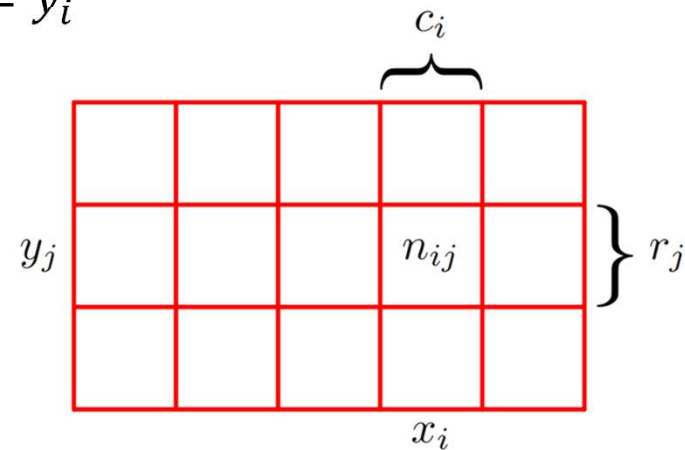
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Joint Probability Distributions

- The probability that $X = x_i$ and $Y = y_j$ is written as:

$$p(X = x_i, Y = y_j)$$

- This is called the **joint probability distribution**



- For an infinite number N of trials, it is given by the proportion of hits in cell i, j

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

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Sum Rule

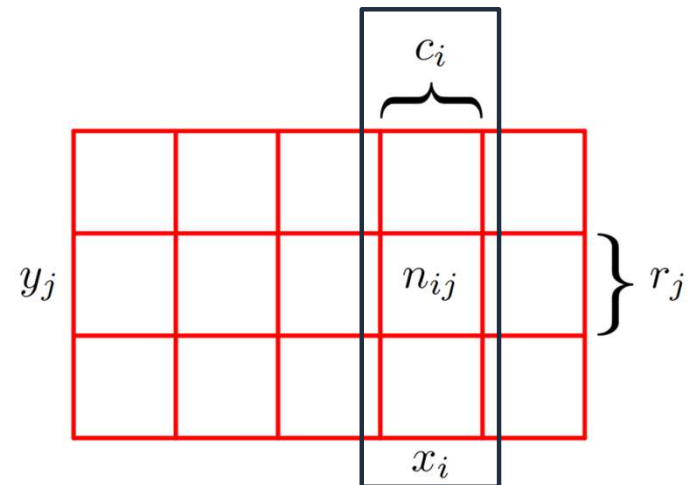
- The probability that X takes the value x_i irrespective of the value of Y is written as $p(X = x_i)$ and is given by the fraction of hits in column i

$$p(X = x_i) = \frac{c_i}{N}$$

- Since c_i is just the sum of the cells of the column, we can also write:

$$p(X = x_i) = \sum_{j=1}^L p(X = x_i, Y = y_j)$$

Note that $p(X = x_i)$ is sometimes called the **marginal probability**, because it is obtained by marginalizing, or summing out, the other variables (in this case Y).



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Product Rule or Conditional Probabilities

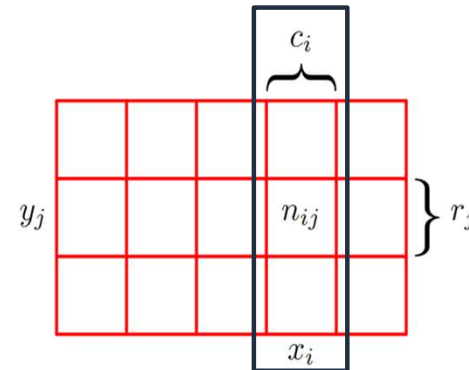
- Now, let's only consider those cases, where $X = x_i$
- We are now interested in the probability that $Y = y_j$ if X is already fixed to x_i
 - E.g., “what is the probability of sampling an apple, after we selected the red box”

- We write this as a **conditional probability**:

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

- With the results from before, we get:

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i} \cdot \frac{c_i}{N} = p(Y = y_j | X = x_i) p(X = x_i)$$



➤ Taken from Christopher M. Bishop, "Pattern Recognition and Machine Learning"

Rules of Probability

- **Sum Rule:**

$$p(X) = \sum_Y p(X, Y)$$

- **Product Rule:**

$$p(X, Y) = p(Y|X)p(X)$$

(**Note the more compact notation:** We simply write $p(B)$ to denote a distribution over the random variable B , i.e., for any value of B , or $p(b)$ to denote the distribution evaluated for the particular value $B = b$, provided that the interpretation is clear from the context.)

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Bayes Theorem

- From the product rule, we can directly derive **Bayes Theorem**:

$$\overbrace{P(Y|X)}^{\text{posterior}} = \frac{\overbrace{P(X|Y)}^{\text{likelihood}} \overbrace{P(Y)}^{\text{prior}}}{\underbrace{P(X)}_{\text{evidence}}}$$

- Prior**: Uncertainty (belief/assumption) about Y before observing X
- Evidence**: The uncertainty about X (regardless of Y)
- Likelihood**: Probability of observing X given that Y is true
- Posterior**: The uncertainty about Y after we have observed X

Random Variables

- A **random variable** X is a variable that can take on different values in a non-deterministic way (i.e., involving *randomness*)
- On its own, a random variable is just a description of the **states** (specific values the variable can take) that are possible
- It must be coupled with a **probability distribution** that specifies how likely each of these states are
- Random variables can be **discrete** or **continuous**

Probability Distributions

- A **probability distribution** is a description of how *likely* each of the possible *states* of a *random variable* (or a set of such variables) is
- The way to describe the distribution depends on whether it is *discrete* or *continuous*

Probability Mass Functions

- The probability distribution over *discrete* variables (e.g., coin tossing, dice rolling) is given by a **probability mass function**
- PMFs are denoted by P and inferred from their argument, e.g., $P(X)$, $P(Y)$
- They can act on many variables, which is known as a joint distribution, written as $P(X, Y)$
- To be a PMF it must satisfy:
 - Domain of P is the set of all possible states x of X
 - $\forall x \in X, 0 \leq P(x) \leq 1$
 - $\sum_{x \in X} P(x) = 1$

Probability Density Functions

- When working with *continuous* variables (e.g., temperature in Odense), we describe probability distributions using **probability density functions**
- To be a pdf, a function p must satisfy:
 - The domain of p must be the set of all possible states x of X
 - $\forall x \in X, p(x) \geq 0$
 - $\int p(x)dx = 1$
- Note, there is no requirement for $p(x) \leq 1$
 - $p(x)$ at a value x is technically NOT a probability, but rather a **likelihood**
 - The area under the curve $p(x)$ for $a \leq x \leq b$ is the probability $P(a \leq X \leq b)$

Conditional and Joint Probability Distributions

- Any probability distribution over many variables (joint dist.) can be decomposed into conditional distributions over only one variable
- For example, applying the product rule successively:

$$P(A, B, C) = P(A, B|C)P(C) = P(A|B, C)P(B|C)P(C)$$

- In the general case:

$$P(X^{(1)}, \dots, X^{(n)}) = P(X^{(1)}) \prod_{i=2}^n P(X^{(i)} | X^{(1)}, \dots, X^{(i-1)})$$

- This holds true for both probability mass (P) and prob. density (p)

Independence

- Two random variables X and Y are **independent** if their probability distribution can be expressed as a product of the marginals:

$$\forall x \in X, y \in Y: p(X = x, Y = y) = p(X = x) \cdot p(Y = y)$$

- This holds true for both probability density (p) and prob. mass (P)

$$P(X, Y) = P(X) \cdot P(Y)$$

- Example: rolling two dice...

Conditional Distributions and Independence

- Recall the product rule: $P(X, Y) = P(Y|X)P(X) = P(X|Y)P(Y)$

- We can rewrite it as: $P(Y|X) = \frac{P(X, Y)}{P(X)}$ and $P(X|Y) = \frac{P(X, Y)}{P(Y)}$

- A direct consequence of **independence** between X and Y :

$$P(Y|X) = \frac{P(X, Y)}{P(X)} = \frac{P(X) \cdot P(Y)}{P(X)} = P(Y)$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{P(X) \cdot P(Y)}{P(Y)} = P(X)$$

- This holds true for both probability mass (P) and prob. density (p)
- The interpretation / intuition should be clear...

Expectation

- Expectation or expected value of a random variable X with distribution $P(X)$ is the average or mean value of X
- For discrete variables:

$$E[X] = \sum_{x \in X} x \cdot P(x)$$

- It is the value we would observe on average over an infinite number of observations of the random variable X
- For continuous variables:

$$E[X] = \int x \cdot p(x) \cdot dx$$

Expectation

- Expectation or expected value of a random variable Y that is a function $f(X)$ of another random variable X with distribution $P(X)$ is the average or mean value of $Y = f(X)$
- For discrete variables:

$$E[f(X)] = \sum_{x \in X} f(x) \cdot P(x)$$

- It is the value we would observe on average over an infinite number of observations of the random variable X
- For continuous variables:

$$E[f(X)] = \int f(x) \cdot p(x) \cdot dx$$

Expectation (Basic Properties)

- Expectation of the **sum** of random variables (or functions thereof):

$$E[X + Y] = E[X] + E[Y]$$

- Expectation of the **product** of **INDEPENDENT** random variables (or functions thereof):

$$E[X \cdot Y] = E[X] \cdot E[Y]$$

Variance

- Variance of a random variable (or function thereof):

$$\begin{aligned} \text{Var}(X) &= E[(X - E[X])^2] \\ \text{Var}(f(X)) &= E[(f(X) - E[f(X)])^2] \end{aligned}$$

- Measures, on average, how much the value deviates from the expectation in a squared sense
- Low variance means values cluster around its expectation
- Square root of the variance is the **standard deviation**
 - $\text{std}(X) = \sqrt{\text{Var}(X)}$

Covariance

- Covariance measures how two variables are **linearly** related:
$$\text{Cov}(f(X), g(Y)) = E[(f(X) - E[f(X)])(g(Y) - E[g(Y)])]$$
- Interpretation:
 - High absolute values: both variables simultaneously deviate from their mean
 - Positive: high/low values of $f(x)$ coincide with high/low values of $g(x)$
 - Negative: high/low values of $f(x)$ coincide with low/high values of $g(x)$
 - Low absolute values: high deviation from mean in one variable can be “cancelled out” by low deviation in the other variable and vice-versa
- Covariance & independence are related but not same
 - Zero covariance is *necessary* for independence *but not sufficient*
 - They must not have nonlinear relationship either

PART 2

Essentials on Continuous Probability Distributions

Definition 7.15 (Cumulative Distribution Function)

The probability distribution of a random variable X is given by its (*cumulative*) *distribution function* $F(x)$, where for any $x \in \mathbb{R}$:

$$F(x) = \Pr(X \leq x).$$

We say that a random variable X is *continuous*, if its distribution function $F(x)$ is a continuous function of x .

Note that:

- ▶ For a continuous random variable X , we must have $\Pr(X = x) = 0$ for any specific value of x .
- ▶ As a consequence, we have $\Pr(X \leq x) = \Pr(X < x)$

Definition 7.16 (Probability Density Function)

If there is a function $f(x)$ such that, for all $-\infty < a < \infty$,

$$F(a) = \int_{-\infty}^a f(t)dt,$$

then $f(x)$ is called the *(probability) density function* of $F(x)$.

Where the derivative of F is well-defined, we have

$$f(x) = F'(x).$$

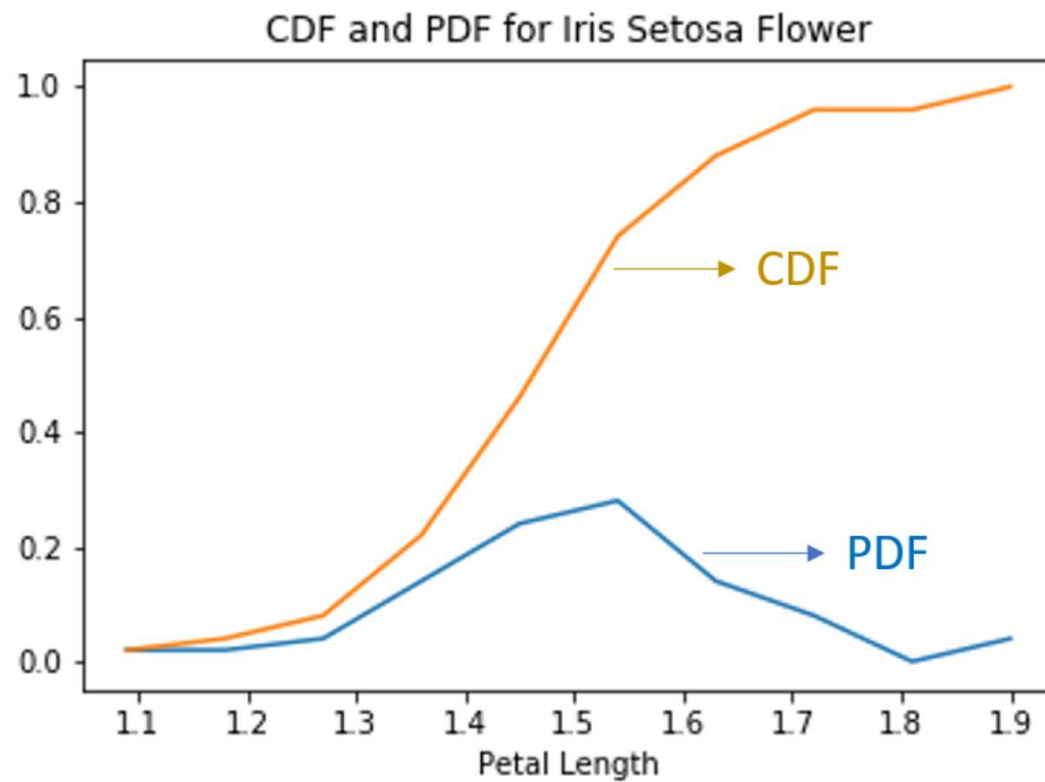
For events in $\mathbb{R}$, the integral is the analogue to the sum of probabilities of simple events (cf. Def. 6.2):

$$\Pr(a \leq X < b) = \Pr(X \in [a, b)) = \int_a^b f(x)dx$$

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Definition 7.17 (Joint Distribution)

The *joint distribution function* of X and Y is
cumulative

$$F(x, y) = \Pr(X \leq x, Y \leq y).$$

The variables X and Y have a joint density function f if, for all x, y :

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv$$

When the derivative exists, we denote

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y).$$

Definition 7.18 (Uniform Distribution)

When a random variable X assumes values in the interval $[a, b]$ such that all subintervals of equal length have equal probability, we say that

- ▶ X has a *uniform distribution* over the interval $[a, b]$
- ▶ X is uniform over the interval $[a, b]$

The uniform distribution is denoted by $\mathcal{U}[a, b]$.

The random variable X following a uniform distribution is denoted by $X \sim \mathcal{U}[a, b]$.

Note that:

Since the probability of taking on any specific value when $b > a$ is 0, the distributions $\mathcal{U}[a, b]$, $\mathcal{U}(a, b)$, $\mathcal{U}[a, b)$, and $\mathcal{U}(a, b]$ are essentially the same.

Probability Distribution and Density Function

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The probability distribution function of a random variable X that is uniform over $[a, b]$ is:

$$F(x) = \begin{cases} 0 & \text{if } x \leq a, \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b, \\ 1 & \text{if } x \geq b \end{cases}$$

and its density function is

$$f(x) = \begin{cases} 0 & \text{if } x < a, \\ \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{if } x > b \end{cases}$$

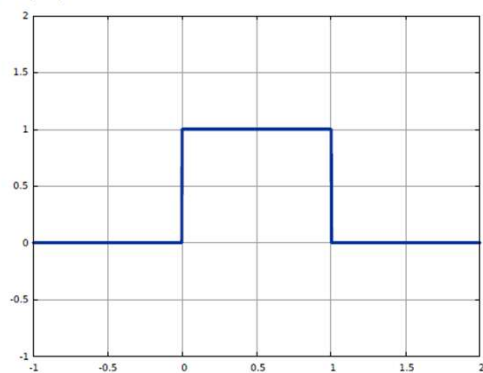
$$E[X] = \frac{b+a}{2}, \quad \text{Var}[X] = \frac{(b-a)^2}{12}$$

Example: $\mathcal{U}[0, 1]$

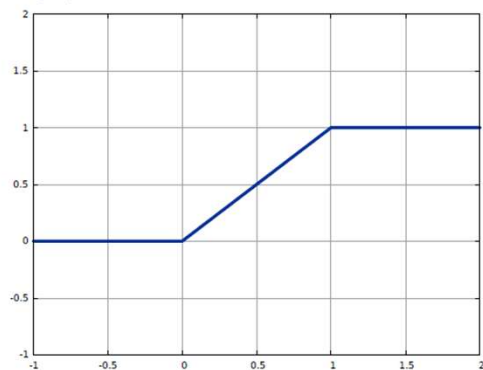
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$f(x)$:



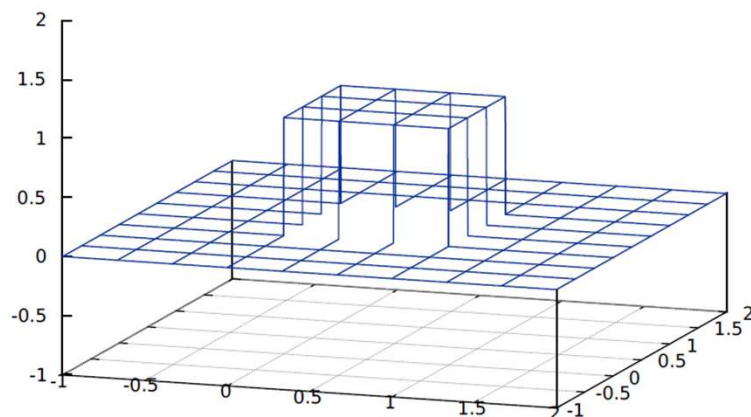
$F(x)$:



Two-dimensional Uniform Distribution

As an example of a joint distribution of two in the interval $[0, 1]$ independently uniformly distributed random variables X and Y , we have a two-dimensional uniform distribution with density function

$$f(x, y) = \begin{cases} 1 & \text{if } (x, y) \in [0, 1]^2 \\ 0 & \text{otherwise} \end{cases}$$



The Normal Distribution

Definition 7.19 (Normal Distribution)

A random variable X follows a *normal distribution* (also: *Gaussian*) with parameters $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}^+$, if the pdf of X for $-\infty < x < \infty$ is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The normal distribution is denoted as $\mathcal{N}(\mu, \sigma^2)$.

Note that:

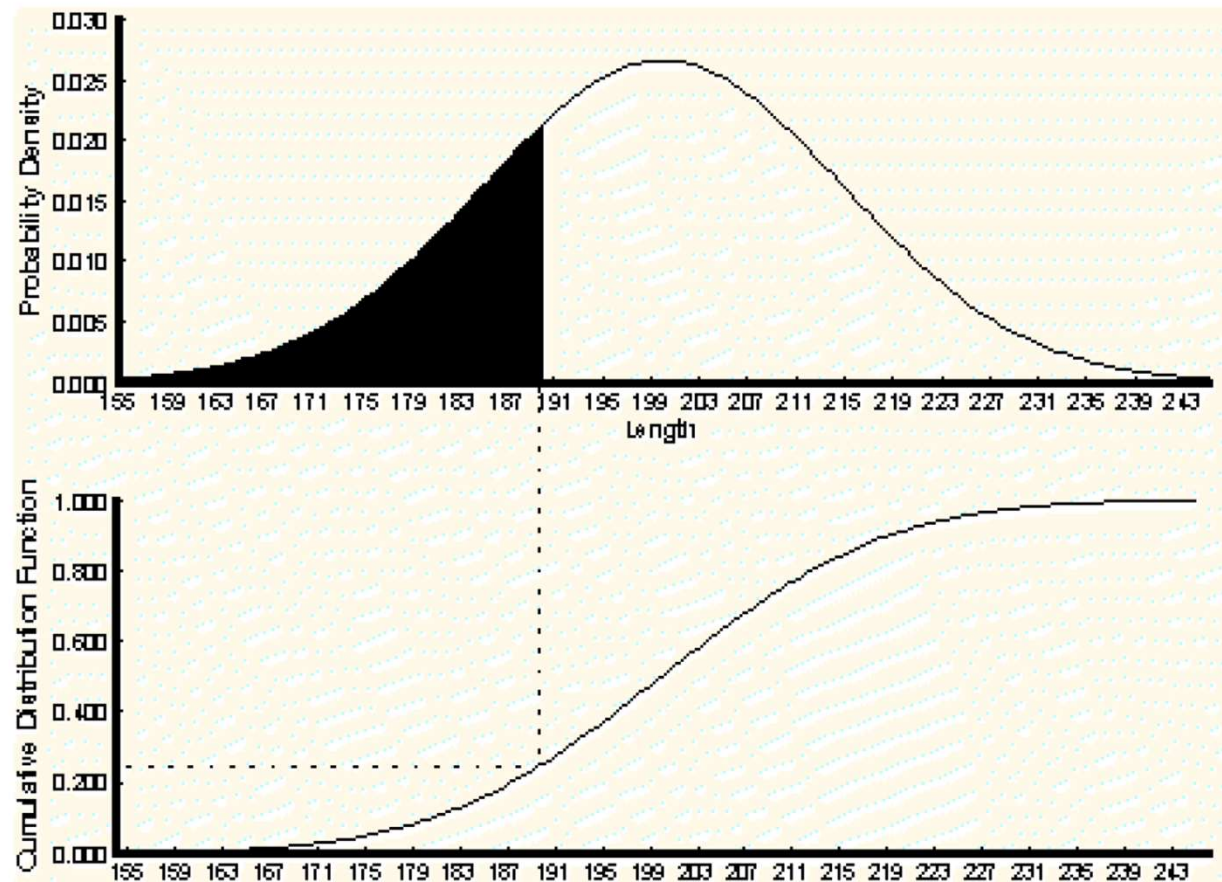
For the parameters μ and σ^2 we have:

$$\mu = E[X] \qquad \sigma^2 = \text{Var}[X]$$

CDF and PDF for some Normal Distribution

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- ▶ If we have a random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, then the random variable

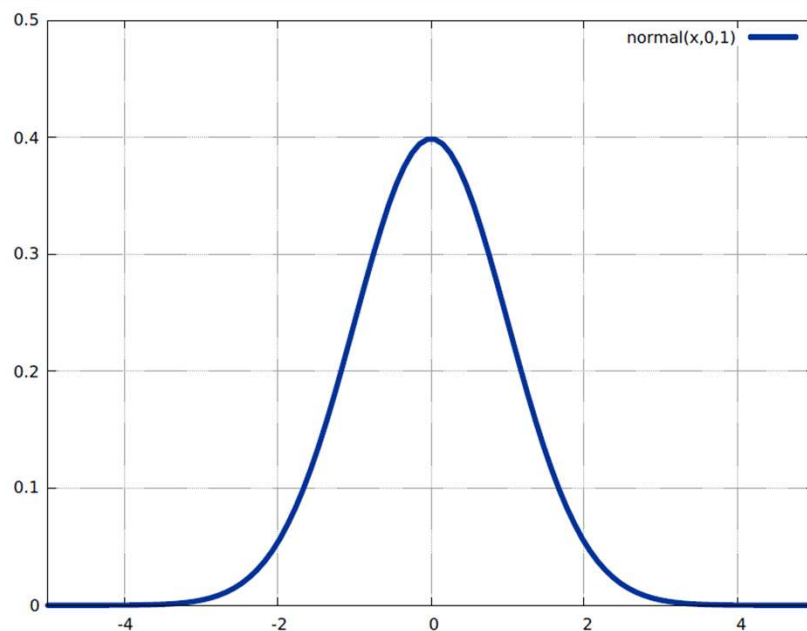
$$Z = \frac{X - \mu}{\sigma}$$

is standard normal distributed.

- ▶ Standardizing a normal distribution is equivalent to expressing the distance from the mean value as some number of standard deviations (sometimes called the z -score).

Definition 7.20 (Standard Normal Distribution)

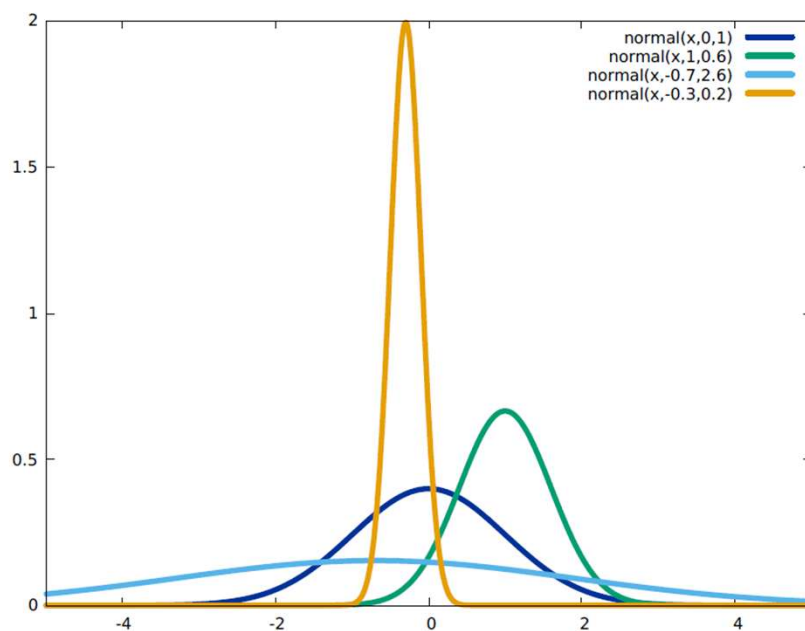
The *standard normal distribution* is given by $\mu = 0$ and $\sigma^2 = 1$, i.e., $\mathcal{N}(0, 1)$.



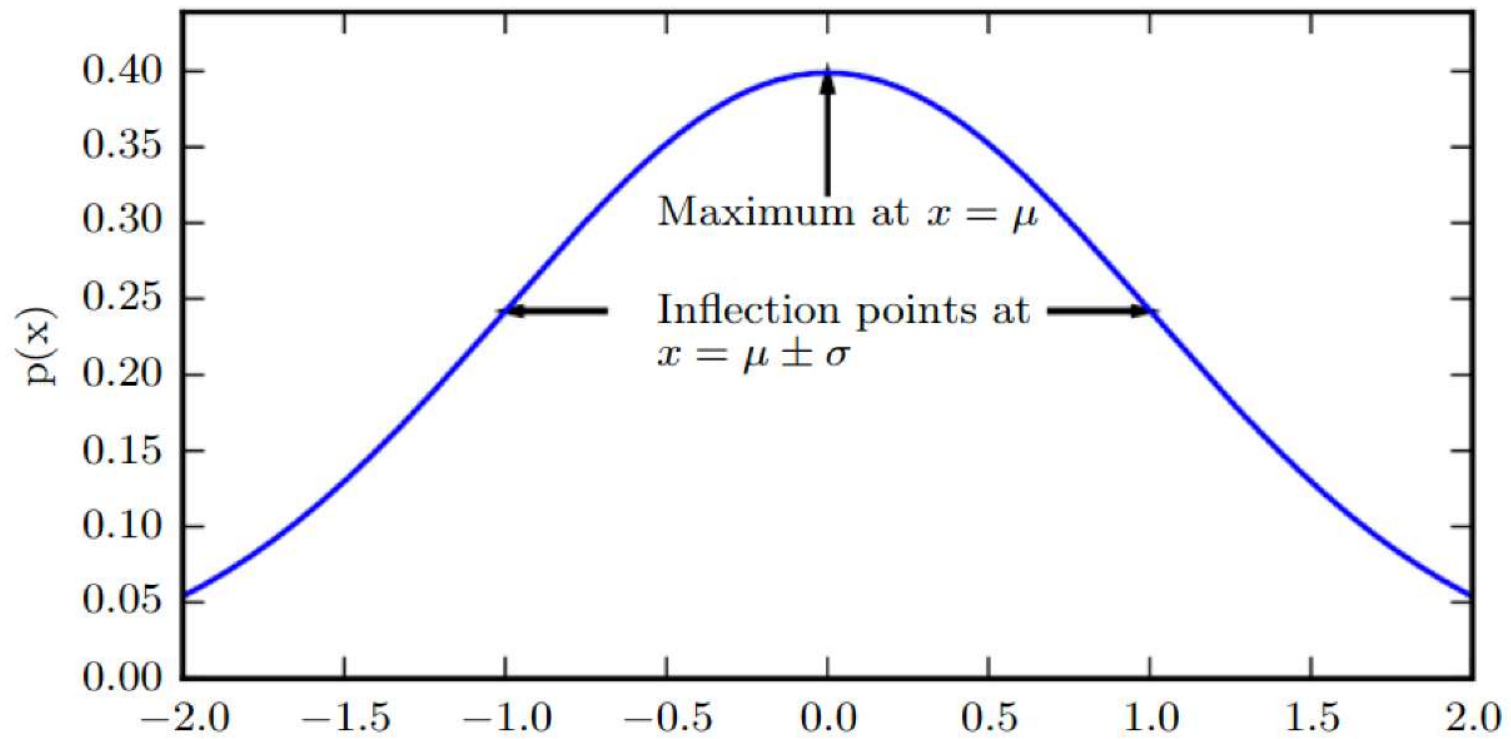
Geometrical Meaning of μ and σ

In general:

- ▶ μ is the center of the symmetric, bell-shaped curve $f(x)$.
- ▶ $\mu + \sigma$ and $\mu - \sigma$ are the inflection points of the curve.



Standard Normal Distribution, $\mu = 0, \sigma = 1$



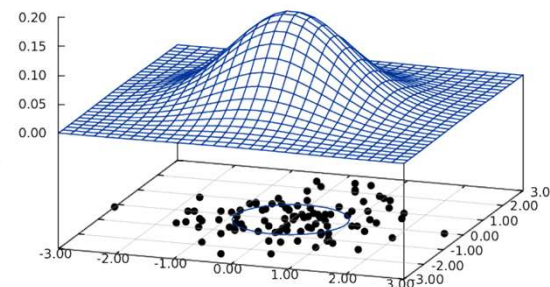
Multivariate Normal Distribution

In the general, d -dimensional case for a vector of normal distributed random variables $\vec{X} = (X_1, \dots, X_d)$, we have the multi-dimensional probability density function for $\vec{x} = (x_1, \dots, x_d)$

$$f(\vec{x}) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \cdot e^{-\frac{(\vec{x} - \vec{\mu}) \Sigma^{-1} (\vec{x} - \vec{\mu})^T}{2}}$$

where $|\Sigma|$ is the determinant of the covariance matrix:

$$\Sigma = \begin{pmatrix} \text{Var}[X_1] & \text{Cov}(X_1, X_2) & \text{Cov}(X_1, X_3) & \cdots & \text{Cov}(X_1, X_d) \\ \text{Cov}(X_2, X_1) & \text{Var}[X_2] & \text{Cov}(X_2, X_3) & \cdots & \text{Cov}(X_2, X_d) \\ \text{Cov}(X_3, X_1) & \text{Cov}(X_3, X_2) & \text{Var}[X_3] & \cdots & \text{Cov}(X_3, X_d) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_d, X_1) & \text{Cov}(X_d, X_2) & \text{Cov}(X_d, X_3) & \cdots & \text{Var}[X_d] \end{pmatrix}$$

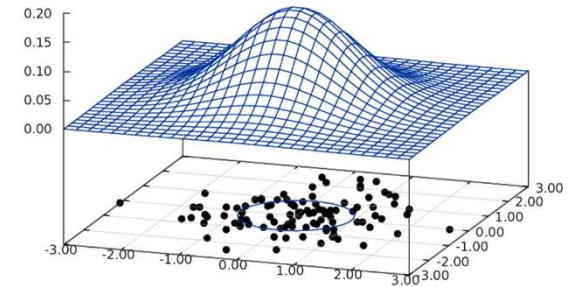


PART 3

Introduction to Density Estimation

Density Estimation

- Assuming that a dataset $X = \{x_1, \dots, x_n\}$, where $x_i \in \mathbb{R}^d$, is a sample from an unknown underlying probability density function $f(x_i) \triangleq f(x_{i1}, \dots, x_{id})$, then **density estimation** refers to the problem of obtaining an estimate $\hat{f}(\cdot)$ of $f(\cdot)$
- There are two major approaches:
 - **Parametric Density Estimation** assumes that a *parametric form* of the distribution $f(\cdot)$ is known, but not its parameters
 - Density estimation then reduces to the simpler problem of estimating from data the parameters of the assumed distribution, which typically can be solved effectively if the assumption holds true
 - For instance, mean and (co)variance(s) of a Normal distribution
 - Different techniques available, e.g.:
 - *Maximum Likelihood Estimation (MLE)*
 - *Method of Moments*

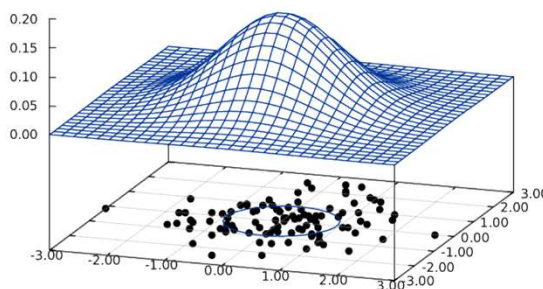


Density Estimation

- Assuming that a dataset $X = \{x_1, \dots, x_n\}$, where $x_i \in \mathcal{R}^d$, is a sample from an unknown underlying probability density function $f(x_i) \triangleq f(x_{i1}, \dots, x_{id})$, then **density estimation** refers to the problem of obtaining an estimate $\hat{f}(\cdot)$ of $f(\cdot)$
- There are two major approaches:
 - **Non-Parametric Density Estimation** is less restrictive as it doesn't make any assumptions about the form/shape of the underlying distribution $f(\cdot)$, which is usually unknown in most data mining applications
 - Different techniques available, most noticeably:
 - ***Kernel-Density Estimation***
 - ***K-Nearest Neighbour (KNN) Density Estimation***

Parametric Density Estimation

- Given a dataset of interest $X = \{x_1, \dots, x_n\}$, if we happen to know that the underlying probability density function $f(x)$ generating the data follows a particular parametric form, then it is usually more effective to obtain an estimate $\hat{f}(\cdot)$ of $f(\cdot)$ by estimating from data only the unknown coefficients (parameters) of $\hat{f}(\cdot)$ restricted to the known parametric form
- One of the most widely used parametric density estimation strategies is MLE:
 - ***Maximum Likelihood Estimation***



Maximum Likelihood Estimation (MLE)

- MLE is one of the most widely used methods to determine the distribution the data at our disposal were sampled from (a.k.a. the *estimation problem*)
- **Parametric MLE:**
 - It is assumed that the observations were sampled from a distribution belonging to a *known parametric family*
 - In other words, the observations $x_1, x_2 \dots x_n$ are independent, identically distributed (i.i.d.) realizations of a random variable X with distribution $f(x | \theta)$
 - either a discrete distribution, a distribution density, or a cumulative distribution
 - The function $f(x_i | \theta)$ treated as a function of the parameter θ with a fixed x_i , is called the **likelihood of observation** x_i
 - The *functional form* of $f(. | .)$ is known, but NOT the value of parameter(s) θ

Maximum Likelihood Estimation (MLE)

- **Goal:** Estimate parameter(s) θ of a probability distribution on the basis of observed realizations of a random variable X
- **Strategy (*MLE Principle*):** Since events with high probability happen more often than those with low probability, then it is natural to assume that what happened was the most likely
 - Best estimate of θ is the value $\hat{\theta}$ that maximizes the likelihood of the i.i.d. sample $x = \{x_1, x_2, \dots, x_n\}$:

$$L(X, \theta) = f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta)$$

$$\hat{\theta} = \arg \max_{\theta} L(X, \theta)$$

Maximum Likelihood Estimation (MLE)

- **MLE (*Log-Likelihood Approach*)**: Mathematically, it is usually easier to work with summations than with products
 - Due to the *monotonicity property of logarithms*, the solution to the original problem is the same as the following problem:

$$\ln L(X, \theta) = \ln f(x_1, x_2, \dots, x_n | \theta) = \sum_{i=1}^n \ln f(x_i | \theta)$$

$$\hat{\theta} = \arg \max_{\theta} \{\ln L(X, \theta)\}$$

Maximum Likelihood Estimation (MLE)

- **Example (*Normal Distribution*):**

- The MLE solution for the Normal distribution is very intuitive, namely: the estimate for the mean is the sample mean of the data, whereas the estimate of the (co)variance is the sample (co)variance
- For instance, in the 1D case:

$$f(x|\theta = \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$
$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

➤ Polanski, A. and Kimmel, M., Bioinformatics, Springer, 2007 (**Section 2.4**)

Non-Parametric Density Estimation

- Next, we will focus on *Non-Parametric Density Estimation* techniques:
 - *Kernel-Density Estimate*
 - *KNN Density Estimate*
- The following material (slides) are from the online website of the textbook:
 - M. J. Zaki and W. Meira Jr., “Data Mining and Machine Learning. Fundamental Concepts and Algorithms”. Cambridge University Press, 2nd edition, 2020
available at https://dataminingbook.info/book_html/
- **Recommended Reading:** Section 15.2 of the above textbook

Univariate Density Estimation

Assume that X is a continuous random variable, and let $x_1, x_2, \dots, x_n$ be a random sample. We directly estimate the cumulative distribution function from the data by counting how many points are less than or equal to x :

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n I(x_i \leq x)$$

where I is an indicator function.

We estimate the density function by taking the derivative of $\hat{F}(x)$

$$\hat{f}(x) = \frac{\hat{F}(x + \frac{h}{2}) - \hat{F}(x - \frac{h}{2})}{h} = \frac{k/n}{h} = \frac{k}{nh}$$

where k is the number of points that lie in the window of width h centered at x . The density estimate is the ratio of the fraction of the points in the window (k/n) to the volume of the window (h).

Kernel Estimator

Kernel density estimation relies on a *kernel function* K that is non-negative, symmetric, and integrates to 1, that is, $K(x) \geq 0$, $K(-x) = K(x)$ for all values x , and $\int K(x)dx = 1$.

Discrete Kernel Define the **discrete kernel** function K , that computes the number of points in a window of width h

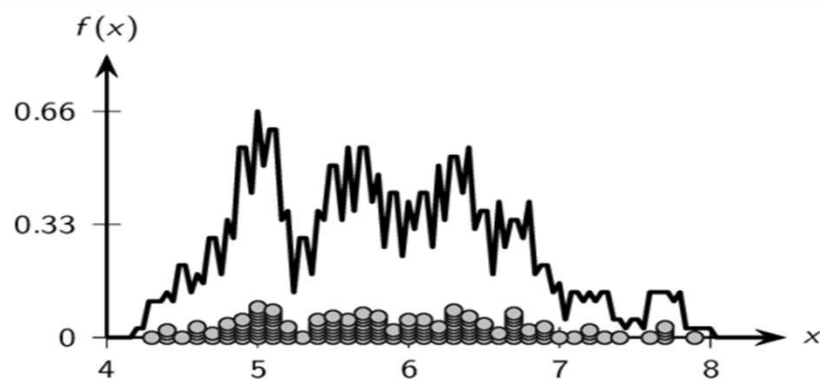
$$K(z) = \begin{cases} 1 & \text{If } |z| \leq \frac{1}{2} \\ 0 & \text{Otherwise} \end{cases}$$

The density estimate $\hat{f}(x)$ can be rewritten in terms of the kernel function as follows:

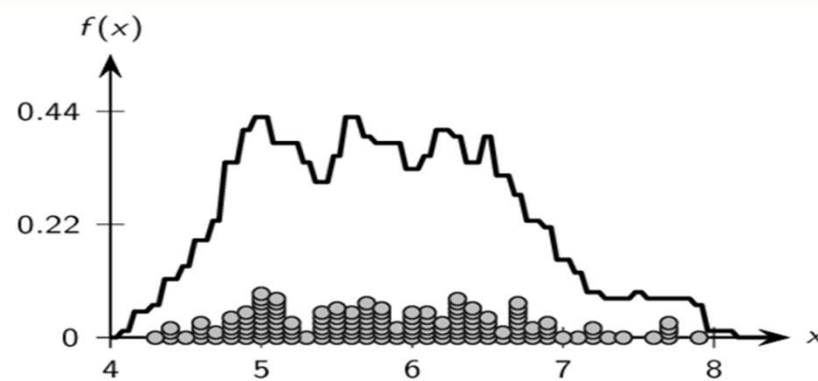
$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

Kernel Density Estimation

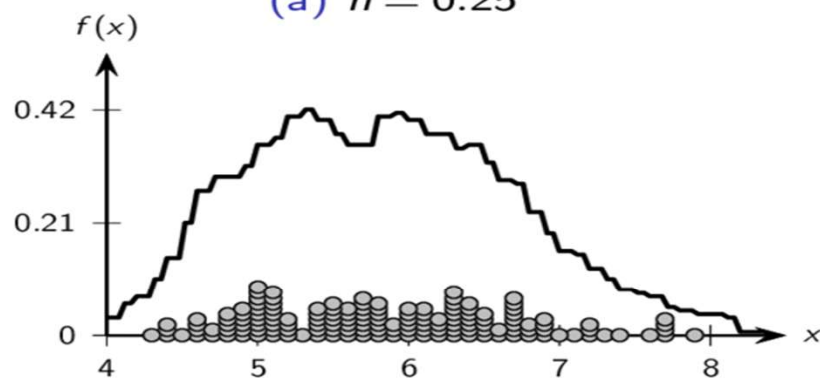
Discrete Kernel (Iris 1D)



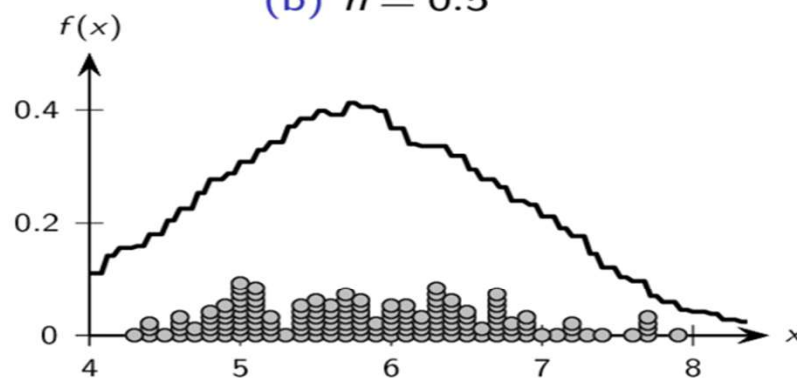
(a) $h = 0.25$



(b) $h = 0.5$



(c) $h = 1.0$



(d) $h = 2.0$

The discrete kernel yields a non-smooth (or jagged) density function.

Kernel Density Estimation

Gaussian Kernel

The width h is a parameter that denotes the spread or smoothness of the density estimate. The discrete kernel function has abrupt changes.

Define a more smooth transition of influence via a Gaussian kernel:

$$K(z) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{z^2}{2} \right\}$$

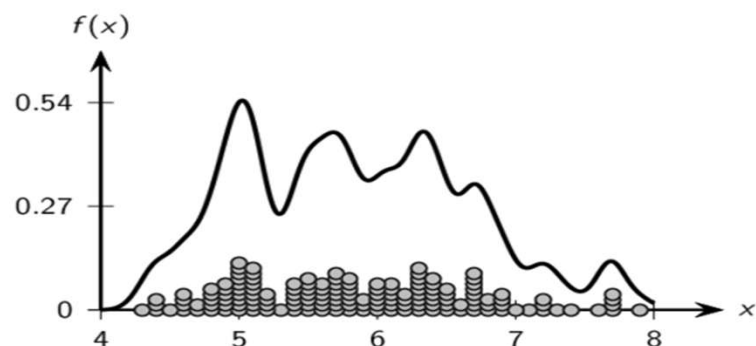
Thus, we have

$$K \left(\frac{x - x_i}{h} \right) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x - x_i)^2}{2h^2} \right\}$$

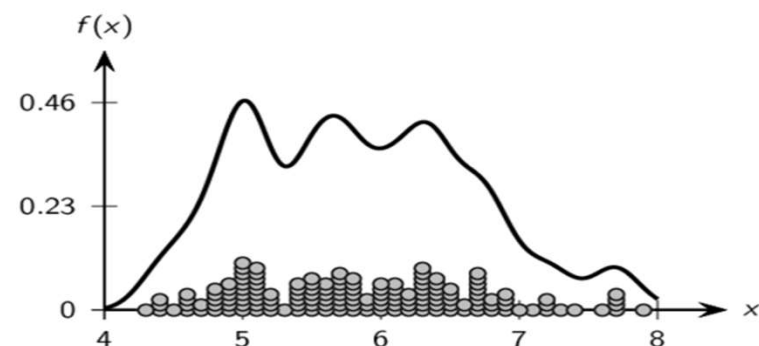
Here x , which is at the center of the window, plays the role of the mean, and h acts as the standard deviation.

Kernel Density Estimation

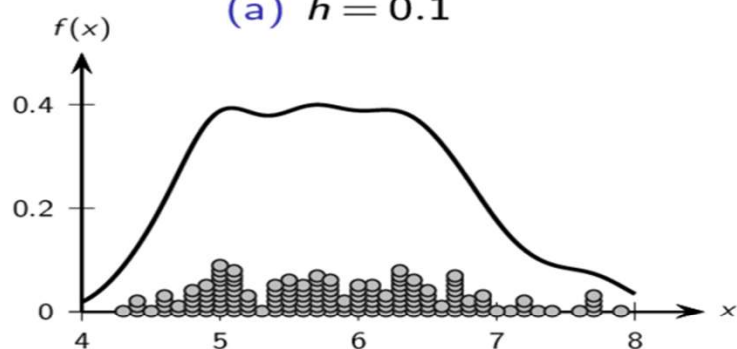
Gaussian Kernel (Iris 1D)



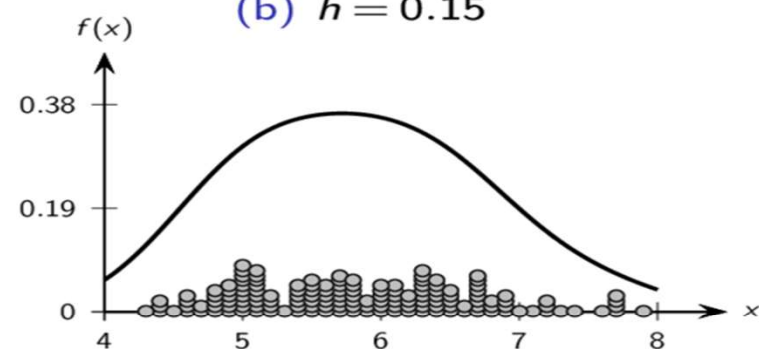
(a) $h = 0.1$



(b) $h = 0.15$



(c) $h = 0.25$



(d) $h = 0.5$

When h is small the density function has many local maxima. A large h results in a unimodal distribution.

Multivariate Density Estimation

To estimate the probability density at a d -dimensional point $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$, we define the d -dimensional “window” as a hypercube in d dimensions, that is, a hypercube centered at $\mathbf{x}$ with edge length h . The volume of such a d -dimensional hypercube is given as

$$\text{vol}(H_d(h)) = h^d$$

The density is estimated as the fraction of the point weight lying within the d -dimensional window centered at $\mathbf{x}$, divided by the volume of the hypercube:

$$\hat{f}(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{\mathbf{x} - \mathbf{x}_i}{h}\right)$$

where the multivariate kernel function K satisfies the condition $\int K(\mathbf{z}) d\mathbf{z} = 1$.

Multivariate Density Estimation

Discrete and Gaussian Kernel

Discrete Kernel: For any d -dimensional vector $\mathbf{z} = (z_1, z_2, \dots, z_d)^T$, the discrete kernel function in d -dimensions is given as

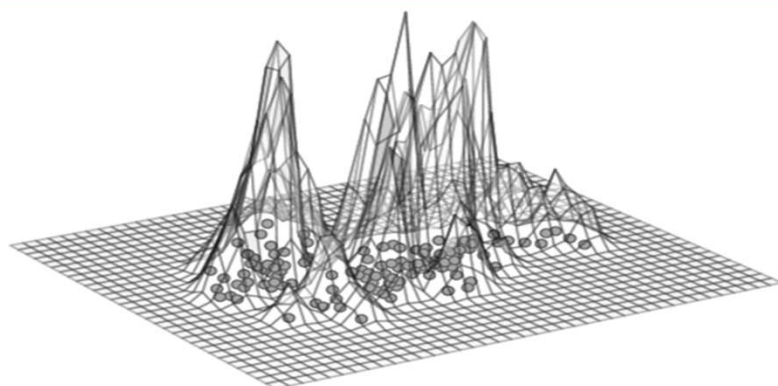
$$K(\mathbf{z}) = \begin{cases} 1 & \text{If } |z_j| \leq \frac{1}{2}, \text{ for all dimensions } j = 1, \dots, d \\ 0 & \text{Otherwise} \end{cases}$$

Gaussian Kernel: The d -dimensional Gaussian kernel is given as

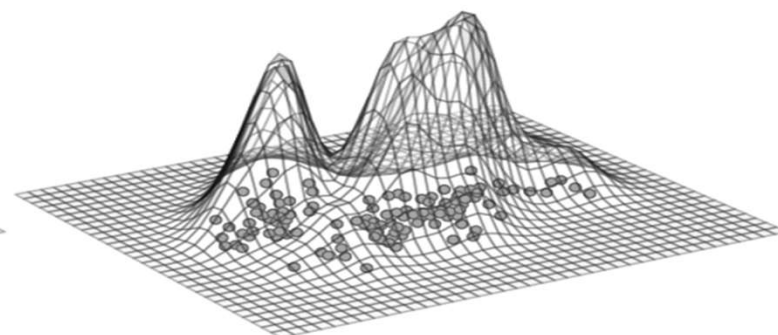
$$K(\mathbf{z}) = \frac{1}{(2\pi)^{d/2}} \exp \left\{ -\frac{\mathbf{z}^T \mathbf{z}}{2} \right\}$$

Density Estimation

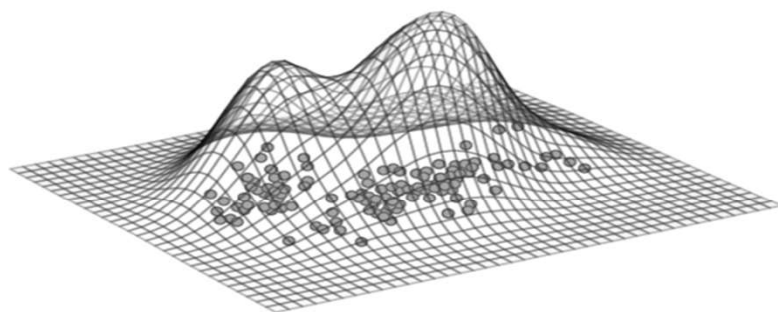
Iris 2D Data (Gaussian Kernel)



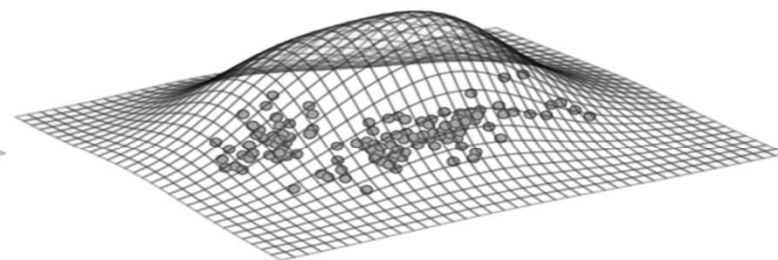
(a) $h = 0.1$



(b) $h = 0.2$



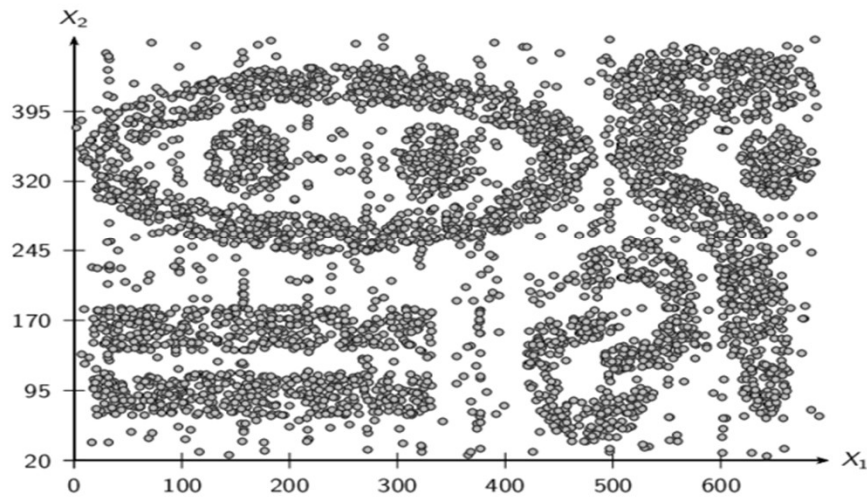
(c) $h = 0.35$



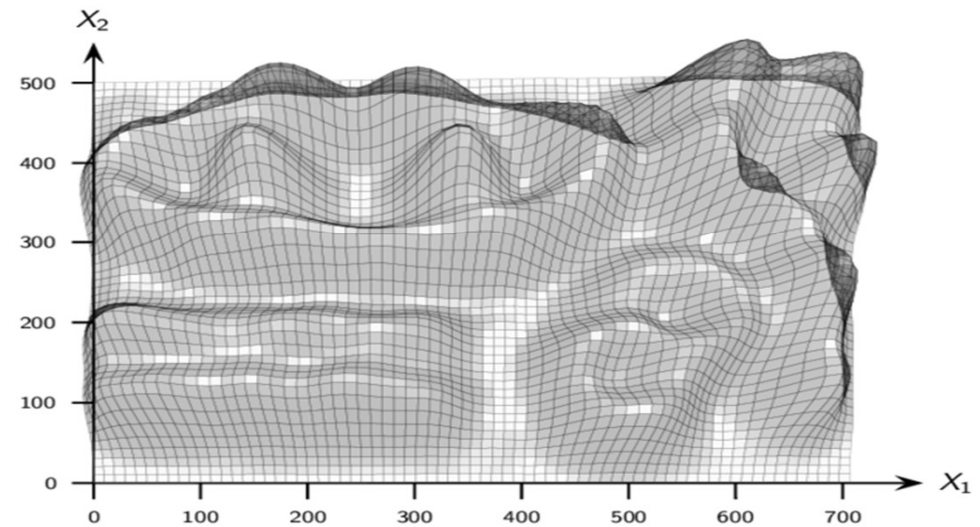
(d) $h = 0.6$

Density Estimation

Gaussian kernel, $h = 20$



(a) Original Points



(b) Gaussian Density Estimation

Nearest Neighbor Density Estimation

In kernel density estimation we implicitly fixed the volume by fixing the width h , and we used the kernel function to find out the number or weight of points that lie inside the fixed volume region.

An alternative approach to density estimation is to fix k , the number of points required to estimate the density, and allow the volume of the enclosing region to vary to accommodate those k points. This approach is called the k nearest neighbors (KNN) approach to density estimation.

Given k , the number of neighbors, we estimate the density at $\mathbf{x}$ as follows:

$$\hat{f}(\mathbf{x}) = \frac{k}{n \text{vol}(S_d(h_{\mathbf{x}}))}$$

where $h_{\mathbf{x}}$ is the distance from $\mathbf{x}$ to its k th nearest neighbor, and $\text{vol}(S_d(h_{\mathbf{x}}))$ is the volume of the d -dimensional hypersphere $S_d(h_{\mathbf{x}})$ centered at $\mathbf{x}$, with radius $h_{\mathbf{x}}$.

hypersphere assuming Euclidean distance (alternatively, hypercube or hyper-diamond for Suprema or Manhattan distances, respectively)

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- M. Mitzenmacher and E. Upfal, “Probability and Computing. Randomization and Probabilistic Techniques in Algorithms and Data Analysis”. Cambridge University Press, 2nd edition, 2017
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